



SEA QUAKE ALARM:

Early Warning For Coastal Safety

PROJECT GUIDE:

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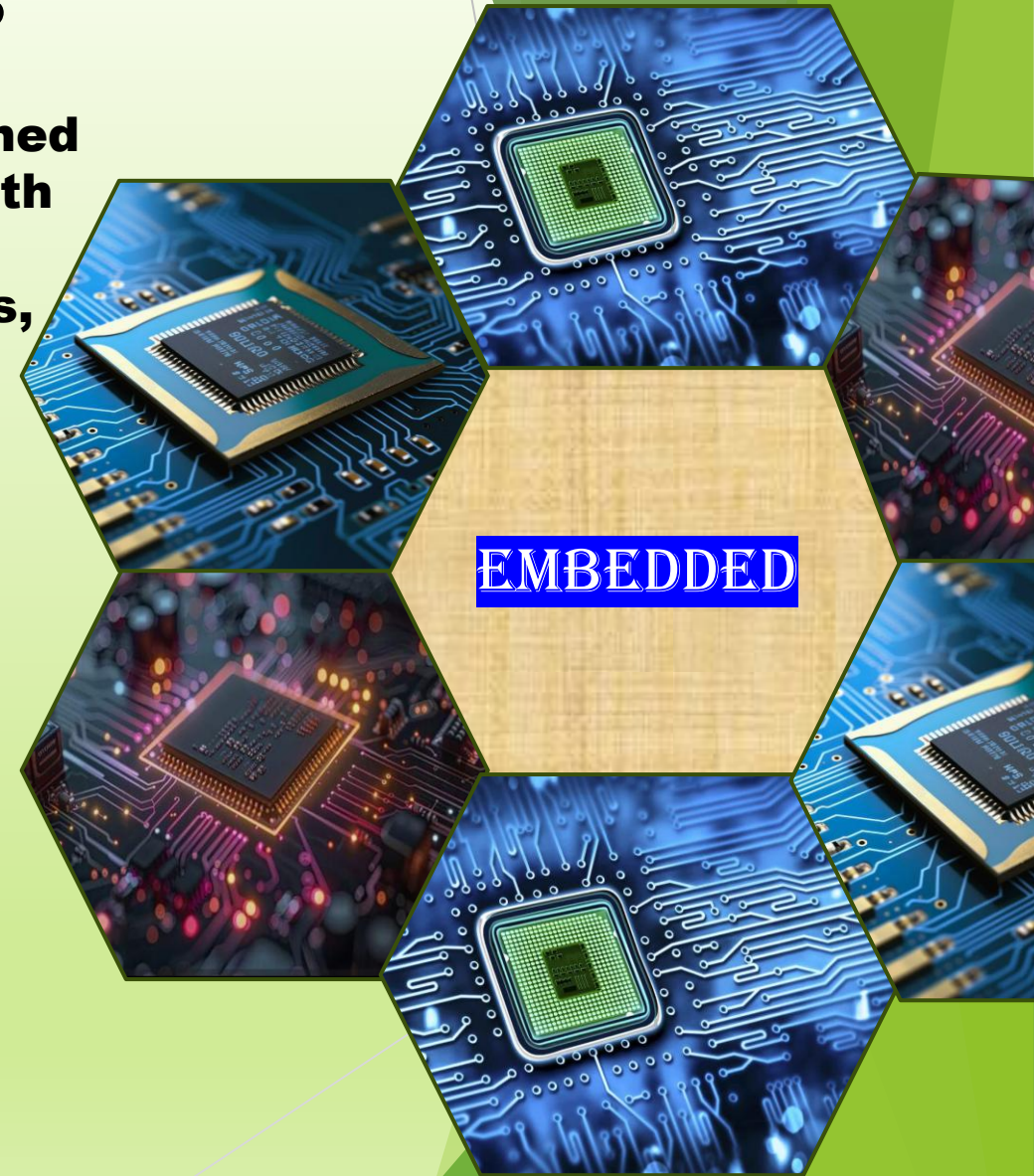
INTRODUCTION TO EMBEDDED SYSTEMS

An Electronic/Electro mechanical system which is designed to perform a specific function and is a combination of both hardware and firmware (Software)

E.g. Electronic Toys, Mobile Handsets, Washing Machines, Air Conditioners

Embedded Systems are:

- ☐ **Unique in character and behavior**
- ☐ **With specialized hardware and software**



#AIM

- ▶ The aim of performing a seaquake alarm for coastal areas safety is to mitigate the devastating impacts of tsunamis and other hazards caused by underwater seismic activity.
- ▶ 🌊 Save lives: This is the primary goal. By providing timely warnings, a seaquake alarm system allows coastal populations to evacuate to higher ground, reducing the risk of casualties.
- ▶ 🌊 Reduce injuries: Early warnings enable people to take protective measures.
- ▶ 🌊 Minimize property damage: With advance notice, communities can secure emergency supplies, and potentially move property to safer locations.

**SEAQUAKE
ALARM**



**EARLY WARNING
FOR COASTAL
SAFETY**

Required Components:-

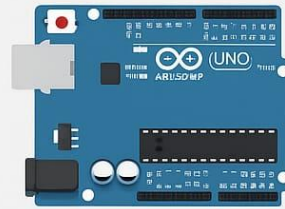
HARDWARE COMPONENTS:

- ▶ 1. Arduino Uno
- ▶ 2. Potentiometer
- ▶ 3. LED (1 piece)
- ▶ 4. Piezo Buzzer
- ▶ 5. Resistors(220Ω or 330Ω).
- ▶ 6. Breadboard
- ▶ 7. Jumper Wires
- ▶ 8. USB Cable (Type B)

SOFTWARE COMPONENTS

- ▶ 1. Arduino IDE
- ▶ 2. Serial Monitor

HARDWARE REQUIREMENTS:



Arduino Uno



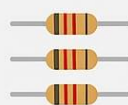
Potene-
meter



LED
(1 piece)



Piezo
Buzzer



Resistors



Jumper
Wires



Breadboard



USB Cable
(Type B)

SOFTWARE REQUIREMENTS:



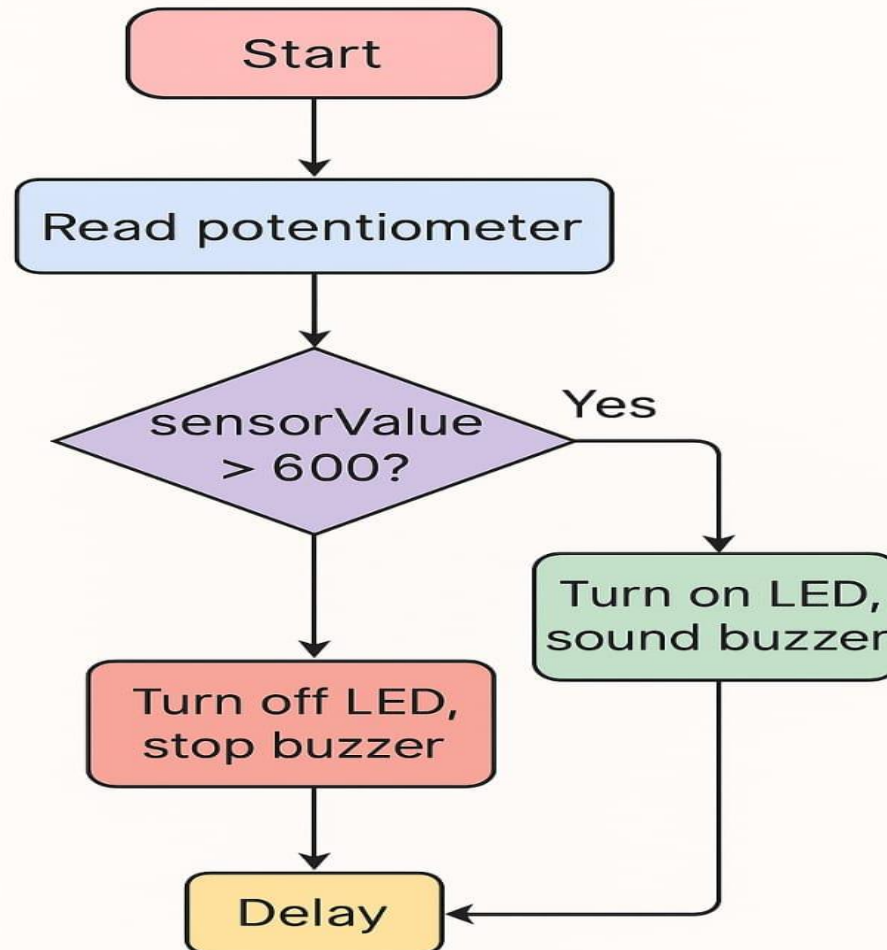
Arduino IDE



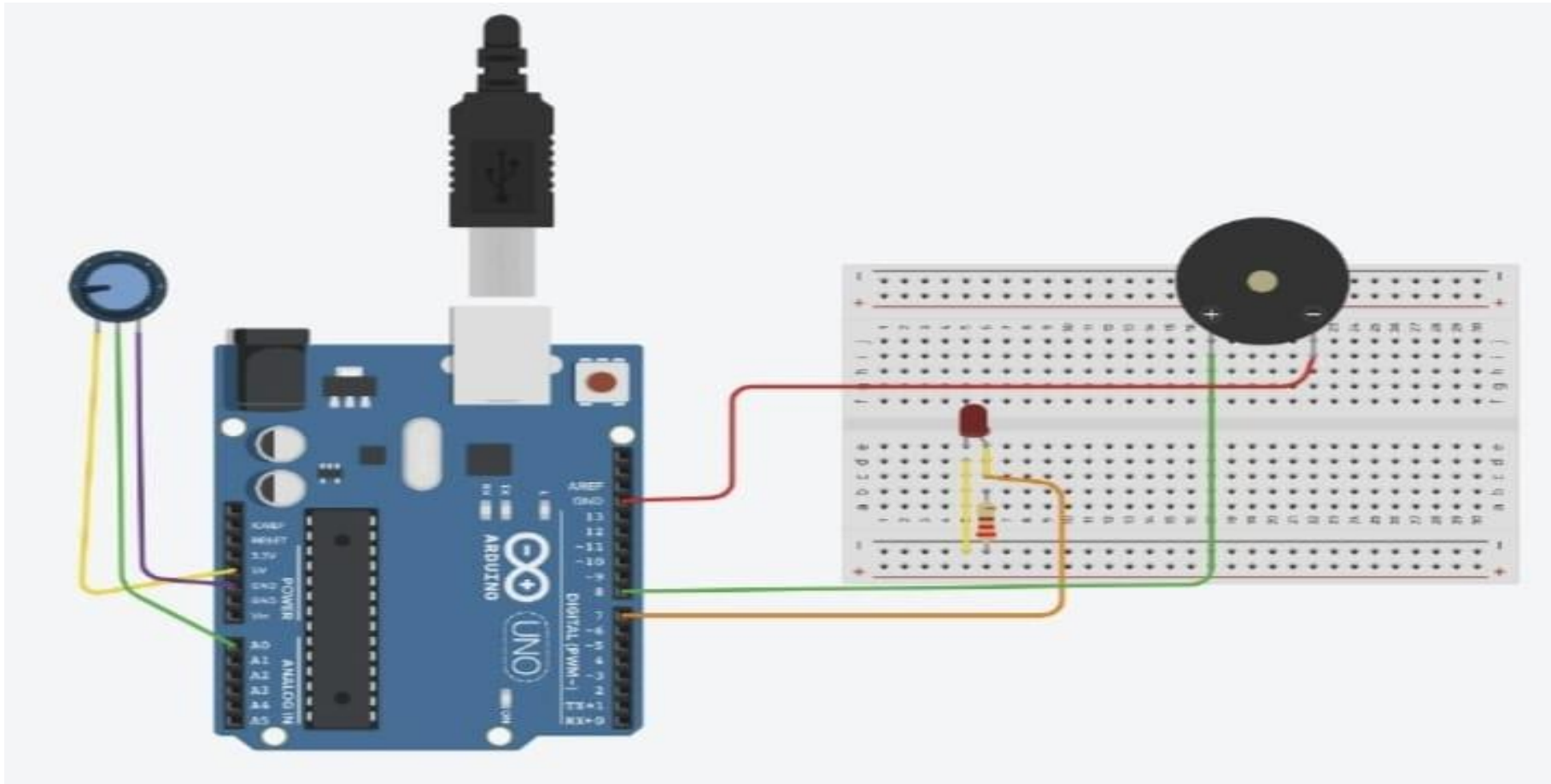
Serial Monitor

- **FLOW CHART**

SEAQUAKE ALARM



- Circuit Diagram



- Arduino Code

```
1  const int potPin = A0;          // Potentiometer (simulated vibration sensor)
2  const int ledPin = 7;           // LED pin
3  const int buzzerPin = 8;        // Piezo buzzer pin
4
5  void setup() {
6    pinMode(ledPin, OUTPUT);
7    pinMode(buzzerPin, OUTPUT);
8    Serial.begin(9600);           // For debugging (optional)
9  }
10
11 void loop() {
12   int sensorValue = analogRead(potPin); // Read potentiometer value (0-1023)
13   Serial.println(sensorValue);          // See value in Serial Monitor
14
15   if (sensorValue > 600) {              // Earthquake threshold
16     digitalWrite(ledPin, HIGH);         // Turn ON LED
17     tone(buzzerPin, 1000);              // Play tone at 1kHz
18   } else {
19     digitalWrite(ledPin, LOW);          // Turn OFF LED
20     noTone(buzzerPin);                  // Stop tone
21   }
22
23   delay(200); // Delay for stability
24 }
```

• OUTPUT

TINKERCAD Incredible Hillar-Bojo

All changes saved

Code Start Simulation Send To

Start / stop simulation Spacebar

```
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Serial Monitor

34° Search ENG IN 15:15 10-06-2025

• Advantages

- ▶ 1. ⚠️ **Saves Lives**
- ▶ 2. 🏥 **Prepares Emergency Services**
- ▶ 3. 🏥 **Prepares Emergency Services**
- ▶ 4. 🏠 **Reduces Property Damage.**
- ▶ 5. 📡 **Supports Disaster Management**
- ▶ 6. 🔄 **Real-time Monitoring & Data Collection**
- ▶ 7. 🌐 **Promotes Community Awareness**



Future Enhancements of the Seaquake Alarm System

- ▶ 1.  **Integrate IoT (Internet of Things)**
- ▶ 2.  **Use Machine Learning for Prediction**
- ▶ 3.  **Solar Powered and Waterproof Design**
- ▶ 4.  **Multiple Sensor Integration**
- ▶ 5.  **GPS Location Tagging**
- ▶ 6.  **Link with Disaster Control Centers**



Conclusion

- ▶ THIS PROJECT HIGHLIGHTS THE IMPORTANCE OF EARLY WARNING SYSTEMS IN REDUCING THE RISK OF NATURAL DISASTERS. BY PROVIDING REAL-TIME ALERTS, IT ENABLES QUICK EVACUATION,
- ▶ 💡 “IN THE FUTURE, THIS PROJECT CAN EVOLVE INTO A REAL-TIME, AI-BASED, SOLAR-POWERED, CLOUD-CONNECTED SMART SYSTEM FOR NATIONAL-LEVEL DISASTER MANAGEMENT.
- ▶ “EARLY DETECTION SAVES LIVES – THIS PROJECT IS A STEP TOWARD SAFER COASTAL LIVING.

THANK YOU